

CAR PARKING SYSTEM – DATABASE LAB

EXPERIMENT NO. 12

TITLE

SIMPLE PROGRAMMING USING WHILE AND REPEAT

AIM

To learn and execute MySQL WHILE and REPEAT loops for repeated operations in a Car Parking System.

DESCRIPTION

WHILE checks its condition before execution. REPEAT executes the statements first and then checks the condition. Stored procedures are used to demonstrate both loops.

STEP 1 – CREATE DATABASE

Create the database.

```
CREATE DATABASE carpark_loops;  
USE carpark_loops;
```

PROGRAM 1 – WHILE LOOP

Build a parking slot string until the value becomes greater than 5.

```
DELIMITER //
```

```
CREATE PROCEDURE parking_while()  
BEGIN  
    DECLARE n INT DEFAULT 1;  
    DECLARE result_text VARCHAR(100) DEFAULT '';  
  
    WHILE n <= 5 DO  
        SET result_text = CONCAT(result_text, 'P', n, ' ');  
        SET n = n + 1;  
    END WHILE;  
  
    SELECT result_text AS parking_slots;  
END //
```

```
DELIMITER ;
```

```
CALL parking_while();
```

PROGRAM 2 – REPEAT LOOP

Increase parking income until it reaches 4000.

DELIMITER //

```
CREATE PROCEDURE parking_repeat()
BEGIN
    DECLARE income INT DEFAULT 1000;

    REPEAT
        SET income = income + 1000;
    UNTIL income >= 4000
    END REPEAT;

    SELECT income AS total_income;
END //
```

DELIMITER ;

CALL parking_repeat();

RESULT

The simple programming exercises using WHILE and REPEAT were executed successfully for the Car Parking System.